



NATIONAL 5 ENGINEERING SCIENCE · MECHANISMS AND STRUCTURES

Materials

Course notes

Version: 27 September 2026

Name _____

Class _____

Learning outcomes

Tick each box when you feel confident you can do it. Come back to this page before your unit test.

BY THE END OF THIS TOPIC...

✓

- | | | |
|---|---|--------------------------|
| 1 | I can classify materials as metals, polymers, ceramics or composites. | ☐ |
| 2 | I can describe material properties such as strength, hardness, toughness, ductility and elasticity. | ☐ |
| 3 | I can select a suitable material for a product and justify my choice using its properties. | ☐ |
| 4 | I can calculate stress using $\sigma = F/A$, including finding the area of a circular cross-section. | ☐ |
| 5 | I can rearrange the stress formula to find force or area. | ☐ |
| 6 | I can calculate strain using $\epsilon = \Delta l/l$, including changing units. | ☐ |
| 7 | I can rearrange the strain formula to find the change in length or the original length. | ☐ |
| 8 | I can use stress and strain calculations to decide whether a design meets its specification. | ☐ |

Where to write your answers: if you have access to a device, use the online course notes — you can type your answers into the answer panels and they are saved as you go. Otherwise, write your answers in your jotter, with the task and question number beside each answer.



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course notes

Section 1: Material Categories

Engineers must choose the most appropriate material for every part of a design. The choice depends on what the part needs to do, what forces it will experience, and the environment it will operate in. Materials are grouped into four main categories:

Metals

- **Steel** — structural beams, car bodies, tools
- **Aluminium** — aircraft, packaging, bike frames
- **Copper** — electrical wiring, plumbing
- **Titanium** — aerospace, medical implants

Polymers (plastics)

- **Polyethylene (PE)** — plastic bags, bottles
- **Polypropylene (PP)** — food containers, rope
- **Nylon** — gears, bearings, textiles
- **Acrylic** — signs, windows, displays

Ceramics

- **Glass** — windows, lenses, optical fibre
- **Concrete** — foundations, roads, bridges
- **Porcelain** — electrical insulators, tiles
- **Silicon carbide** — cutting tools, brake discs

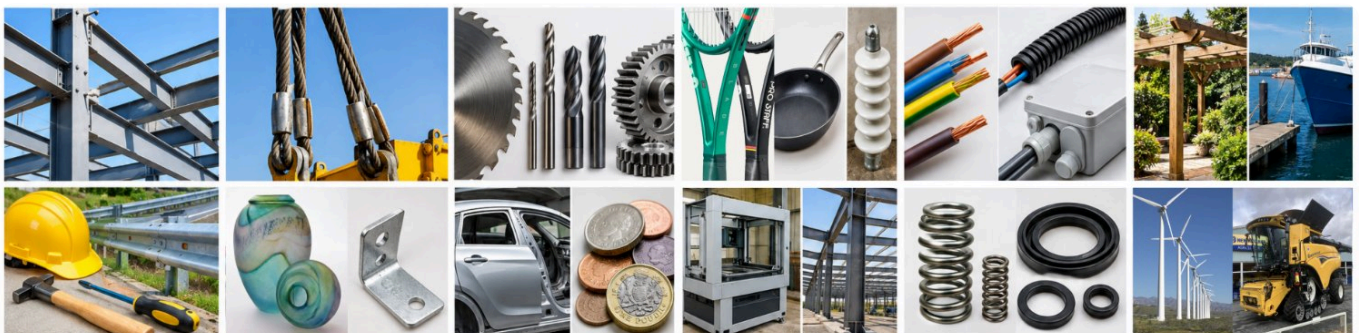
Composites

- **Carbon fibre reinforced polymer (CFRP)** — aircraft, F1 cars, bikes
- **Glass fibre (GRP/fibreglass)** — boat hulls, helmets
- **Reinforced concrete** — buildings, bridges

Section 2: Material Properties

To choose the right material, engineers consider a range of **mechanical, physical and other properties**. Understanding these properties allows you to justify a material choice in the exam.

Property	Definition	Example where important
Strength	Ability to withstand a force without breaking	Structural beams, lifting cables
Hardness	Resistance to scratching or indentation	Cutting tools, drill bits, gear teeth
Toughness	Ability to absorb impact energy without fracturing	Helmets, crash barriers, tool handles
Ductility	Ability to be drawn into wires or bent without breaking	Electrical wiring, bent brackets
Malleability	Ability to be hammered or pressed into shape	Aluminium car body panels, coins
Stiffness (rigidity)	Resistance to bending or deformation under load	Machine frames, structural columns
Elasticity	Ability to return to original shape after a force is removed	Springs, rubber seals
Density	Mass per unit volume — how heavy the material is	Aircraft components — low density preferred
Thermal conductivity	How well the material conducts heat	Heat sinks, cooking pans, insulators
Electrical conductivity	How well the material conducts electricity	Wiring (conductors), casings (insulators)
Corrosion resistance	Resistance to chemical attack or rusting	Outdoor structures, marine applications
Cost	Material and processing cost per unit	Always a factor in real engineering decisions



Task 1 — Material categories and properties

1. State which category (metal, polymer, ceramic or composite) each of the following belongs to: (a) copper, (b) nylon, (c) concrete, (d) carbon fibre reinforced polymer, (e) glass.
2. Explain the difference between strength and toughness. Give an example of a material that is strong but brittle (not tough).
3. State two properties that are important when selecting a material for electrical wiring. For each property, explain why it matters.

Section 3: Selecting a Material

In the SQA exam you will be given a scenario and asked to select a suitable material and **justify your choice**. A good justification always links the properties of the material directly to the demands of the application.

A complete material selection answer always has two parts: **(1) name the material** and **(2) state the specific property or properties that make it suitable**, linked clearly to the application.

How to approach a material selection question

1. Read the scenario carefully — identify what forces or conditions the part must withstand
2. List the key requirements (e.g. must be lightweight, must conduct electricity, must resist corrosion)
3. Match a material to those requirements
4. Justify each property you mention — say **why** it matters for this specific application

WORKED EXAMPLE 1 — BICYCLE FRAME

Question: A bicycle frame must be lightweight and strong. Select a suitable material and justify your choice.

GOOD ANSWER

Aluminium alloy would be a suitable material. It has a **high strength-to-weight ratio**, meaning the frame can withstand the forces from the rider without being unnecessarily heavy. Aluminium also has good **corrosion resistance**, which is important as bicycle frames are used outdoors in wet conditions. Its **ductility** allows it to be formed into the required tube shapes during manufacture.

WORKED EXAMPLE 2 — ELECTRICAL CABLE INSULATION

Question: Select a suitable material for the plastic insulation surrounding electrical cables and justify your choice.

✓ GOOD ANSWER

PVC (polyvinyl chloride) is commonly used. It is an **electrical insulator**, which prevents the user from receiving an electric shock by stopping current flowing through the outer casing. It is also **flexible**, so the cable can be bent without cracking, and it is **low cost** making it economical to produce in large quantities.

WORKED EXAMPLE 3 — BRIDGE SUPPORT CABLE

Question: A suspension bridge cable must carry very large tensile forces and operate outdoors for many years. Select a suitable material and justify your choice.

✓ GOOD ANSWER

High-tensile steel wire is used. Steel has very high **tensile strength**, meaning it can carry the enormous pulling forces in the cables without breaking. It is also **stiff**, which limits the amount of sag in the bridge. Galvanised or stainless steel variants are used to provide **corrosion resistance** for long outdoor service life.

Task 2 — Material selection and justification

1. A cutting tool is used to machine hard metals in a factory. Select a suitable material for the cutting tool and justify your choice by referring to at least two properties.

2. A designer is choosing a material for the body panels of a sports car. The panels must be lightweight, strong and able to be formed into curved shapes. Suggest a suitable material and justify your answer.

3. A food packaging company needs a material for a disposable drinks bottle. The material must be lightweight, transparent, cheap to produce and safe for food contact. Select a suitable material and justify your choice.

4. An engineer is designing a cooking pan. The base must conduct heat efficiently and be strong enough to survive being dropped. The handle must not conduct heat. Identify suitable materials for (a) the pan base and (b) the handle. Justify both choices.

5. A wind turbine blade must be very stiff, lightweight and able to withstand continuous cyclic loading over many years. Suggest a suitable material and justify your choice.

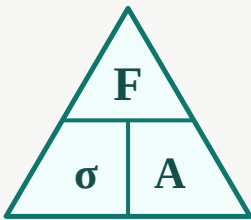
Section 4: Stress

When a force is applied to a structural member, the material inside it is being **squeezed or stretched**. The intensity of this internal force — the force per unit area — is called **stress**. Stress is what determines whether a material will fail: if the stress in a member exceeds the material's strength, it will break.

$$\text{Stress } (\sigma) = \frac{\text{Force (N)}}{\text{Area (mm}^2\text{)}}$$

Stress is measured in N/mm² (also written as MPa). The symbol is the Greek letter sigma (σ).

Rearrangements: Force = Stress $\times$ Area | Area = $\frac{\text{Force}}{\text{Stress}}$

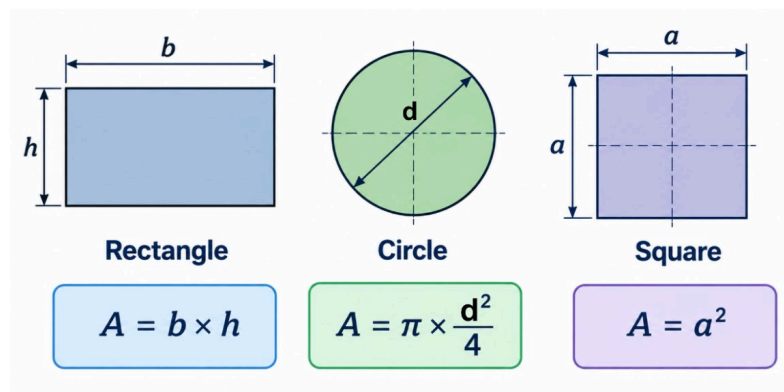


FσA formula triangle. Cover up the quantity you want to find. If the two that are left sit **side by side**, multiply them. If one sits **above** the other, divide the top by the bottom. Cover σ and you see F over A; cover F and you see σ next to A.

The **cross-sectional area** is the area of the cut face of the member — perpendicular to the direction of the force. A smaller cross-section experiences **greater stress** for the same force. This is why load-bearing columns and cables are sized carefully.

Stress tells us how hard the material is being pushed or pulled internally. A high stress value means the material is close to its limit — choose a stronger material, or increase the cross-sectional area.

Calculating cross-sectional areas



WORKED EXAMPLE 1 — STRESS IN A SQUARE BAR

A square bar of 20 mm × 20 mm cross-section is subjected to a tensile load of 500 N. Calculate the stress in the bar.

1

Write the formula: $\sigma = \frac{F}{A}$

2

Calculate area: $A = 20 \times 20 = 400 \text{ mm}^2$

3

Calculate stress: $\sigma = \frac{500}{400} = 1.25 \text{ N/mm}^2$

WORKED EXAMPLE 2 — MAXIMUM LOAD ON A COLUMN

A column of cross-section 0.25 m^2 has a maximum allowable stress of 50 N/mm^2 . Calculate the maximum compressive load.

1

Convert area to mm^2 : $0.25 \text{ m}^2 = 0.25 \times 10^6 \text{ mm}^2 = 250\,000 \text{ mm}^2$

2

Rearrange: $F = \sigma \times A = 50 \times 250\,000 = 12\,500\,000 \text{ N} = 12.5 \text{ MN}$

WORKED EXAMPLE 3 — MINIMUM WIRE DIAMETER

A steel wire supports a load of 8 kN. The stress must not exceed 200 N/mm^2 . Find the minimum wire diameter required.

1

Convert load: $8 \text{ kN} = 8\,000 \text{ N}$

2

Find minimum area: $A = \frac{F}{\sigma} = \frac{8\,000}{200} = 40 \text{ mm}^2$

3

Find diameter from circular area formula: $A = \frac{\pi d^2}{4}$

$$d^2 = \frac{4A}{\pi} = \frac{4 \times 40}{\pi} = 50.93$$

$$d = \sqrt{50.93} = 7.14 \text{ mm}$$

Task 3 — Stress calculations

Show all working. Check units carefully — forces in N, areas in mm², stress in N/mm².

1. A rectangular steel bar has a cross-sectional area of 250 mm² and is subjected to a force of 50 kN. Determine the stress in the bar.
2. A wire 4 mm in diameter is subjected to a force of 300 N. Find the stress in the wire.
3. What diameter of round steel bar is required to carry a tensile force of 10 kN if the stress must not exceed 14.14 N/mm²?
4. A piece of wire 0.75 mm in diameter is subjected to a force of 120 N. Calculate the stress.
5. A mild steel tie-bar of circular cross-section has a diameter of 30 mm and is subjected to a tensile pull of 75 kN. Find the tensile stress in the bar.
6. A concrete column has a square cross-section of 200 mm × 200 mm and supports a compressive load of 500 kN. Calculate the compressive stress. Would a material with an allowable stress of 12 N/mm² be suitable? Justify your answer.

Section 5: Strain

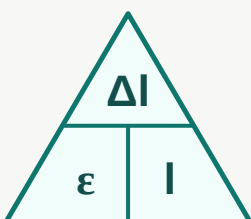
Every material deforms slightly when a force is applied to it — it stretches under tension or shortens under compression. This change in length, expressed as a **proportion of the original length**, is called **strain**. Strain tells us how much a material has deformed relative to its size.

$$\text{Strain } (\epsilon) = \frac{\text{Change in length } (\Delta l)}{\text{Original length } (l)}$$

Strain has **no units** — it is a dimensionless ratio. The symbol is the Greek letter epsilon (ϵ).

Length (l) is measured in **metres (m)**. Δl and l must be in the **same units** before dividing.

Rearrangements: $\Delta l = \epsilon \times l$ and $l = \frac{\Delta l}{\epsilon}$



$\Delta l \epsilon l$ formula triangle. It works exactly like the stress triangle: cover the quantity you want. Cover ϵ and you see Δl over l ; cover Δl and you see ϵ next to l , so multiply.

Strain is always a very small number — often expressed in scientific notation (e.g. 5×10^{-4}). It has no units because it is a ratio of two lengths.

Metres or millimetres?

Length (l) is measured in **metres (m)**, so the original length of a member is usually given in m. The change in length (Δl), however, is often tiny and is usually given in **mm**.

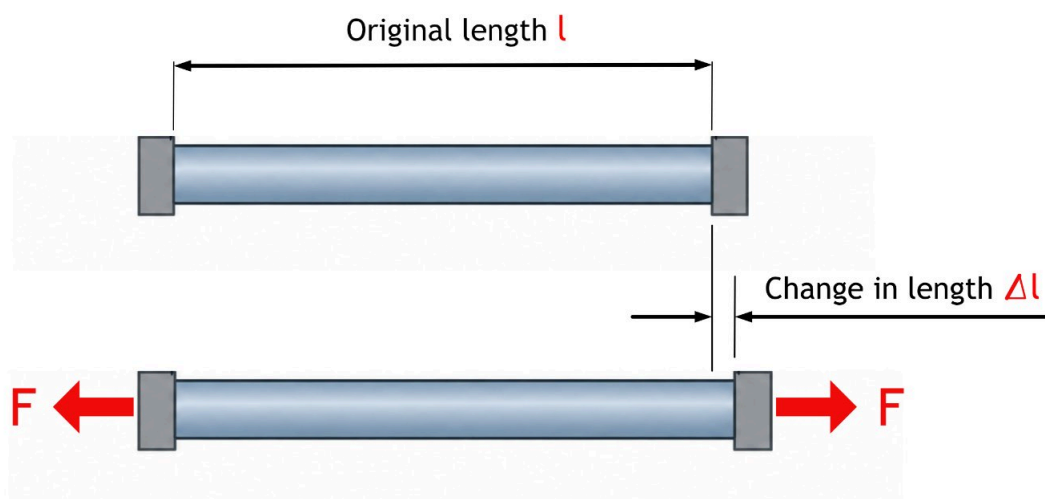
Because strain is one length divided by another, the units **cancel out**. **As long as Δl and l are in the same unit, the calculation works** — you get exactly the same answer whether you use metres or millimetres.

When the change in length is a small decimal in mm (for example 0.06 mm), converting it to metres gives a number full of zeros (0.00006 m), and it is very easy to lose or add one. In that case it is often **less confusing to convert the original length to mm** instead, and work entirely in millimetres.

Pick one unit and convert **both** lengths into it before dividing. Working in m or in mm gives the same strain — choose whichever keeps the numbers simplest.

Tensile and compressive strain

If a member is under **tension** (being stretched), Δl is an **increase** in length — the strain is positive. If a member is under **compression** (being shortened), Δl is a **decrease** in length — the strain is sometimes described as compressive.



WORKED EXAMPLE 1 — STRAIN IN A WIRE

A steel wire of length 5 m stretches 2.5 mm when a load is applied. Calculate the strain.

1

Write the formula: $\epsilon = \frac{\Delta l}{l}$

2

Method A — work in metres: $l = 5 \text{ m}$, $\Delta l = 2.5 \text{ mm} = 0.0025 \text{ m}$

$$\epsilon = \frac{0.0025}{5} = 0.0005$$

3

Method B — work in millimetres: $l = 5 \text{ m} = 5\,000 \text{ mm}$, $\Delta l = 2.5 \text{ mm}$

$$\epsilon = \frac{2.5}{5\,000} = 0.0005$$

4

Answer: both methods give $\epsilon = 0.0005 (= 5 \times 10^{-4})$. Strain has no units.

WORKED EXAMPLE 2 — FINDING CHANGE IN LENGTH

The allowable strain on a concrete column must not exceed 5×10^{-4} . The column is 3 m tall. Find the maximum reduction in height.

1

Rearrange: $\Delta l = \epsilon \times l$

2

Work in metres: $l = 3 \text{ m}$

3

$$\Delta l = 5 \times 10^{-4} \times 3 = 0.0015 \text{ m} = \mathbf{1.5 \text{ mm}}$$

WORKED EXAMPLE 3 — COMBINED STRESS AND STRAIN

A mass of 2 500 kg is hung from a vertical steel bar. The bar has a rectangular cross-section of 75 mm × 50 mm and an original length of 0.5 m. The bar stretches by 2.5 mm. Calculate both the stress and the strain. (Use $g = 9.81 \text{ m/s}^2$.)

1

Find the force: $F = mg = 2\,500 \times 9.81 = 24\,525 \text{ N}$

2

Find the area: $A = 75 \times 50 = 3\,750 \text{ mm}^2$

3

$$\text{Calculate stress: } \sigma = \frac{F}{A} = \frac{24\,525}{3\,750} = \mathbf{6.54 \text{ N/mm}^2}$$

4

Calculate strain (working in mm keeps the numbers simple): $l = 0.5 \text{ m} = 500 \text{ mm}$, $\Delta l = 2.5 \text{ mm}$

$$\epsilon = \frac{2.5}{500} = \mathbf{0.005}$$



Task 4 — Strain calculations

Show all working. Remember: Δl and l must be in the same unit before dividing — if the change in length is a small decimal in mm, it is often easier to convert l to mm. State clearly that strain has no units.

1. A wire 10 m long stretches 5 mm when a force is applied. Find the strain.

2. A tow bar 1.5 m long is compressed by 4.5 mm during braking. Find the strain.

3. The allowable strain on a bar is 0.0075 and its length is 2.5 m. Find the maximum change in length.

4. During testing, a shaft was compressed by 0.06 mm and the resulting strain was 0.00012. What was the original length of the shaft?

5. A mild steel tie-bar, 5 m long with a 30 mm diameter, lengthens by 1.5 mm under a tensile pull of 75 kN. Calculate both the stress and the strain in the bar.

6. An aluminium alloy rod (cross-section 40 mm $\times$ 20 mm, length 0.8 m) is subjected to a compressive force of 64 kN and compresses by 0.4 mm. Calculate the stress and strain. Would a material with an allowable stress of 80 N/mm² be suitable? Justify your answer.

Section 6: Design Challenges

These challenges bring everything together: choosing a material, justifying it against a specification, sizing a part using stress, and checking a prototype using strain. They follow the same shape as an assignment task — a context, a numbered specification, then design, test and evaluate. Work through each part in order and refer back to the specification by number when you justify or evaluate.

Design Challenge 1: Café sign hanger rods

Design, test and evaluate

A café on a seafront promenade wants a heavy wooden sign to hang from a bracket above its door. The sign will hang from **two round rods**, each 0.4 m long, which will be in **tension**. You have been asked to choose a material for the rods and work out what size they need to be. The rods must meet the following specification:

- i Each rod must carry a tensile force of **1 200 N**.
- ii The stress in each rod must not exceed **60 N/mm²**.
- iii The rods must survive in salty, damp sea air for at least 10 years without significant corrosion.
- iv The strain in each rod must not exceed **0.0004**.

(a) Select a suitable material for the rods. Justify your choice, referring to specification point iii and at least one other property.

(b) Calculate the minimum diameter of rod that meets specification ii. Rods are only available in whole-millimetre diameters — state which size you would use.

(c) A prototype rod of the size you chose in (b) was tested with the full 1 200 N load and stretched by 0.09 mm. Calculate the strain and state whether specification iv is met.

(d) Evaluate your design against each specification point in turn. Suggest one improvement or further test you would make before the sign is installed.

Design Challenge 2: Community garden planter legs

Design, test and evaluate

A community garden is building raised planters at wheelchair height so that everyone can reach the soil. Each planter stands on **four square legs**, each 0.6 m tall, which will be in **compression**. When full of wet soil, the load on

each leg is 6 000 N. The legs must meet the following specification:

- i Each leg must carry a compressive force of **6 000 N**.
- ii Each leg must have a square cross-section no bigger than **100 mm × 100 mm** so it fits the frame.
- iii The legs must not rot or corrode while standing on damp ground for many years.
- iv The legs should be made from recycled material wherever possible.
- v Each leg must not shorten by more than **0.3 mm** under full load.

The garden has three materials it could use. The allowable compressive stress is the highest stress each material should be designed to carry.

Material	Allowable compressive stress	Notes
Softwood timber	7 N/mm ²	Cheap and easy to cut; rots in damp ground unless treated
Recycled plastic lumber (HDPE)	2 N/mm ²	Made from 100% recycled plastic; does not rot
Glass fibre composite (GRP)	40 N/mm ²	Very strong and does not rot; difficult to recycle and expensive

(a) For each of the three materials, calculate the minimum side length of square leg needed to carry the load in specification i.

(b) Select the most suitable material. Justify your choice by referring to each of specifications ii, iii and iv.

(c) A 60 mm × 60 mm prototype leg made from your chosen material was loaded to 6 000 N and shortened by 0.25 mm. Calculate the stress and the strain in the leg.

(d) Evaluate the prototype leg against specifications i, ii and v. Describe one change you could make to the leg if it had failed specification v.